

which integrates an ARM Cortex-A9 MPCore processor on 28nm FPGA, Virtex series and Spartan series of FPGAs. Altera offers the Stratix, Arria and Cyclone series of FPGAs.

Details of I²C protocol, hardware connection and VHDL code are presented in the following sections.

I²C protocol

I²C is a simple bi-directional serial bus, which supports multiple masters and slaves. The I²C protocol was developed by Philips in 1982 to reduce the connection pin count. It comprises a serial bi-directional data line (SDA) and a serial bi-directional clock line (SCL). Both these lines are open drained. SDA must be pulled up to V_{CC} with a 5.6-kilo-ohm resistor and SCL also needs to be pulled up with the same resistor only if there are multiple masters in the system or when the slave will perform clock stretching as a flow control measure to synchronise with the master. I²C bus supports various modes, namely, the slow-speed mode (maximum speed of 10kbps), standard mode (maximum speed of 100kbps), fast mode (maximum speed of 400kbps), fast mode plus (maximum speed of 1Mbps) and

high-speed mode (maximum speed of 3.4Mbps).

Each device connected to the I²C bus is software addressable by a unique address. The device that controls the sending and receiving of messages by controlling the bus access is referred to as the master and the devices that are controlled by the master are referred as slaves. Both the master and the slaves can transmit and receive messages. It is a synchronous protocol that allows a master device to initiate communication with a slave device. The master device controls the SCL, which dictates the timings of all transfers on the I²C bus.

The I²C protocol comprises a set of conditions to establish communication between the devices. These include:

1. Start condition
2. Sending of the slave address along with read/write information by the master
3. Acknowledgement by the slave device
4. Data transmission between the master and the slave device
5. Stop condition

Prior to any transmission, a start condition needs to be issued on the

bus. The start condition acts as a signal to all connected ICs that something is about to be transmitted on the bus. It is initiated by the master by sending a high-to-low transition on the SDA line whilst SCL is high (Fig. 2).

After the start condition, the 7-bit slave address followed by the eighth bit is sent on the SDA line by the master, where a '0' on bit indicates a write operation from the master to the slave (master is the transmitter and slave is the receiver) and a '1' indicates a read operation from the slave to the master (master is the receiver and slave is the transmitter). The MSB of the address bits is sent first followed by the other bits. The value on the SDA line can be changed when SCL line is low.

The slave device whose address is the same as the address being sent out by the master responds with an acknowledgement bit on the SDA line by pulling the SDA line low during the ninth clock cycle of the SCL line. After that, the transmission takes place between the master and the slave. Each byte on the SDA line for transmission is 8 bits long with the MSB being sent first. Also, the SDA line must not be changed when SCL line is high, except for the start and the stop conditions. Any number of bytes can be transmitted, however each byte is to be followed by an acknowledgement bit. Fig. 3 shows the transmission process.

To signal the end of data transfer, the master sends a stop condition by pulling the SDA line from low to high whilst the SCL line is at high (Fig. 4).

I²C master controller implementation

In this section, the implementation of the I²C master controller using an FPGA is discussed. Any FPGA can be chosen as per the designers' requirement. Fig. 5 shows the interconnection between the FPGA and the slave device. As one can see, there are two lines required

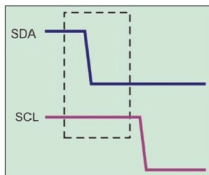


Fig. 2: Start condition

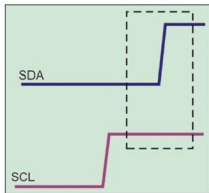


Fig. 4: Stop condition

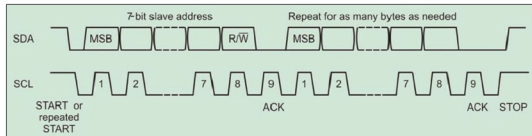


Fig. 3: Data transmission